TKA Protocol Phase 1

Purpose

Protection

Typical Time Frame

0-2 Weeks

Precautions

- WBAT w/ AD
- Monitor wound healing
- Monitor for signs of DVT (Pt should be taking at least 81mg aspirin 2x/day per surgeon prescription)
- Avoid putting pillows or towel rolls under knee to promote extension

Criteria to Pass

- 0°-90° of knee AROM
- Independent transfers and gait with appropriate AD

Priority List & Suggested Exercises

1. **Knee Extension ROM**
   a. Low Load Long Duration: Extension prop, Foot on Ottoman (add weight or ice to all)
   b. Manual Techniques: Passive extension overpressure (add quad set), patella mobs (superior glides)
   c. Quad sets (supine & seated)
   d. Stretches: Hamstring & Slantboard
2. **Knee Flexion ROM**
   a. PROM: Seated knee bend off table, Physioball knee flexion (supine), Heel slides, Wall slides, Standing knee flexion on step
   b. Nu-step and bike (½ revolutions)
   c. Manual Techniques: Patella mobs (inferior glides)
3. **Quad Activation**
   a. Quad sets (pushing into towel roll, 10sec hold) - optional NMES
   b. SLR with NMES (quad set > into SLR > reset) - optional NMES
4. **Gait Training**
   a. Weight shifting with UE assistance
   b. Gait training with RW (heel to toe) > SPC
   c. Calf raises

Special Considerations

- Swelling: compression, elevation, ice, retrograde massage
- Incision: inspect frequently to ensure healing and closing; Aquacel dressing typically removed between 10-14 days post-op/patients can shower with this on
- Pain Management: Ice and elevate regularly, use of prescribed pain meds and education on taking before PT
- Educate patient on expectations to look at progress from week to week rather than day to day
- Most common complaints but very normal: swelling, stiffness following inactivity, and sleeping
- During all transfers encourage knee flexion

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TKA Protocol Phase 2

Purpose

Functional Restoration

Typical Time Frame

2-8 Weeks

Precautions

- Monitor wound healing (avoid scar tissue mobilization until wound closed)
- Monitor for signs of DVT
- Avoid putting pillows or towel rolls under knee to promote extension

Criteria to Pass

- Knee AROM 0°- 120°
- Ambulate >500ft without AD, deviations, or antalgia
- SLR 2x10 without extension lag
- Sit to stand without UE from 20” surface
- Modified Step Up Test: 5 reps, 8" step, UE for balance
- Modified LSD Test: 5 reps, 6” step, toe tap strategy, UE for balance
- TUG <13sec with appropriate AD

Priority List & Suggested Exercises

1. **Knee Extension ROM**
   a. Continue Phase 1 exercises (add sets and reps per discretion)
   b. More aggressive PROM
2. **Knee Flexion ROM**
   a. Continue Phase 1 exercises (add sets and reps per discretion); More aggressive PROM
   b. Distal quad tissue work/cupping
   c. Nu-step & Bike (1/2 revolutions > full)
   d. AAROM: Shuttle Press, TRX sit to stands
3. **Quad Activation**
   a. Quad set into SLR - optional NMES
   b. TKEs (banded, ball behind knee, prone)
   c. SAQ > LAQ - optional NMES
   d. Double leg shuttle press
4. **Gait Training**
   a. Gait training w/ AD > sled pushes > w/o AD
   b. Retro walking, side stepping
   c. Low hurdle (step to pattern > reciprocal pattern)
5. **Functional Strength**
   a. Calf raises
   b. Bridges (focus on = knee flexion)
   c. Sit to stand variations
   d. Modified Step up progression
   e. LSD progression
6. **Balance Training**
   a. Feet Together > Feet together on foam > tandem stance > SLS
   b. Marching on foam > step taps (increasing height)

Special Considerations

- Swelling: compression, elevation, ice, retrograde massage
- Incision: Should be closed by 6 weeks
- Can begin scar tissue mobilization once wound closed
- Collaborate with surgeon and consider manipulation around 6 weeks if patient continues to <100°
- If patient is having difficulty getting past 110° you can still move onto next phase if that is the only criteria remaining
- If patient returning to work during this phase ensure they are aware of frequent position changes, and elevating LE throughout their day
- ****IF PATIENT’S PLOF WAS VERY LOW THEY MAY DC AFTER THIS PHASE****

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TKA Protocol Phase 3

Purpose

Progressive Overload

Typical Time Frame

6-12 Weeks

Precautions

- Avoid sleeping with pillow under knee to promote extension
- Avoid standing or walking for long periods to decrease swelling

Criteria to Pass

- Pain free and 90% ROM of uninvolved knee
- Symmetrical squat to 20" Box
- Reciprocal gait pattern on steps without handrail
- 6MWT match norms for age
- Get up from floor w/ UE assist if needed
- 30sec STS test 17” height: Complete 10 reps
- SLS match norms for age:
  - 30sec <70 y/o, 15sec 70-79 y/o, 10sec >80 y/o

Priority List & Suggested Exercises

1. **Knee Flexion ROM**
   a. Continue Phase 2 exercises (adding sets and reps per discretion)
   b. Manual: prone quad stretch, tissue work to distal quad and/or hamstring
   c. Knee Circles
   d. AAROM: Shuttle press, TRX squat
   e. Standing hamstring activation > high hurdle step overs (reciprocal)
2. **Anterior Chain Strength**
   a. TKA Leg extension progression
   b. Wall sit progression
   c. Heavy banded TKEs/Shuttle Press TKEs
   d. SL Shuttle press
3. **Functional Strength**
   a. Squat progression (partial range)
   b. Step ups progression
   c. LSD > Forward step down progression
   d. Get up from floor progression
3. **Functional Strength (continued)**
   a. Posterior Chain: hinges, HS curls, bridges, calf raises
   b. Farmer carries
   c. Lateral lunges with TRX
4. **Balance Training**
   a. Tandem stance > SLS > SLS w/ KB pass through
   b. 3-way balance reach > star drill
   c. Lateral band walks
5. **Gait Training**
   a. Toe/heel walking
   b. Sled push/pull
   c. Incline treadmill walking
   d. Endurance training (walking, rower, bike, elliptical)

Special Considerations

- Collaborate with surgeon and consider manipulation around 6 weeks if patient continues to struggle with ROM
- Continued ROM programming and HEP to ensure no loss of ROM and improve feeling of stiffness
- Ride the bike 30min per day if stiffness is persisting
- Patient may not always achieve 90% ROM, as some patients struggle more than others
- Should be off of narcotic pain medication by this phase
- If patient returning to work during this phase ensure they are aware of frequent position changes, and elevating LE throughout their day
- ****MOST PATIENTS WILL DC AFTER THIS PHASE****

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TKA Protocol Phase 4

Purpose

High Level Adult

Typical Time Frame

10-16 Weeks

Precautions

- Avoid excessive and repetitive impact exercises

Criteria to Pass

Use Strengthlevel.com norms:

- Functional Squat test: 10 reps w/ 15lb (f), 20lb (m) med ball to 18" bench
- Functional RDL Test: 10 reps w/ 20lb (f) 25lb (m) med ball from 8" step
- Complete 3 minute step test (8") w/o UE support
- 30sec STS test 17” height: Complete 15 reps

Priority List & Suggested Exercises

1. **Anterior Chain Strength**
   a. TKA Leg Extension Progression
   b. Wall Sit Progression
   c. Heavy Banded TKEs/Shuttle Press TKEs
   d. SL Shuttle press > Eccentric Shuttle Press
2. **Functional Strength**
   a. Goblet squats (partial ROM)
   b. Lateral step downs > forward step downs (as needed)
   c. Split squats > Reverse lunges (partial ROM)
   d. Get up from floor progression
   e. Posterior Chain: Hinges, bridges, HS curls, calf raises
   f. Incorporate more compound movements
3. **Core/Glute Strength**
   a. Plank variations
   b. Clamshells > Sidelying abduction
   c. Stork stance
   d. Sidelying shuttle press
   e. Hip Hikes
   f. Lateral Band walks
   g. Lateral Squat progression

Special Considerations

- ****Only highly active individuals will progress to this phase****
- Continued ROM programming and HEP to ensure no loss of ROM and improve feeling of stiffness
- Criteria to pass this phase can also be based on patient’s individual discharge goals
- Be aware of swelling following activity and rest as needed

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